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## Rangkuman Praktikum Sistem Terdesentralisasi dan Terdistribusi

### Membangun dan Menguji REST API antar VM

Instans (1/2) [Info](#)

Hubungkan Status instans Tindakan Luncurkan instans

Temukan instans berdasarkan atribut atau tanda (case-sensitive)

Semua sta...

|                                     | Name      | ID Instans          | Status instans | Tipe instans | Pemeriksaan statu | Status alarm  | Availability Zone | DNS IPv4 publik          | Alamat |
|-------------------------------------|-----------|---------------------|----------------|--------------|-------------------|---------------|-------------------|--------------------------|--------|
| <input type="checkbox"/>            | Vm-server | i-0d42f28a4cd65cf21 | Menjalankan    | t3.micro     | 3/3 pemeriksaan   | Lihat alarm + | us-east-1b        | ec2-34-237-146-178.co... | 34.237 |
| <input checked="" type="checkbox"/> | Vm-client | i-000a8ff1c561f54d3 | Menjalankan    | t3.micro     | 3/3 pemeriksaan   | Lihat alarm + | us-east-1b        | ec2-44-210-41-98.com...  | 44.210 |

EC2 > Grup Keamanan > sg-093719f1f9357fa13 - launch-wizard-4 > Edit aturan masuk

### Edit aturan masuk [Info](#)

Aturan masuk mengontrol lalu lintas masuk yang diizinkan untuk menjangkau instans.

| Aturan masuk         | Jenis             | Protokol | Rentang port | Sumber | Deskripsi - opsional |
|----------------------|-------------------|----------|--------------|--------|----------------------|
| sg-01927254777e2784f | TCP Kustom        | TCP      | 3000         | Kustom | 0.0.0.0/0            |
| sg-02ac2e802c090fe5e | Semua ICMP - IPv4 | ICMP     | Semua        | Kustom | 0.0.0.0/0            |
| sg-09756b4fac08fa674 | SSH               | TCP      | 22           | Kustom | 0.0.0.0/0            |

Tambahkan aturan

Aturan dengan sumber 0.0.0.0/0 atau ::/0 memungkinkan semua alamat IP mengakses instans Anda. Sebaiknya atur aturan grup keamanan untuk mengizinkan akses hanya dari alamat IP yang diketahui.

Batalan Pratinjau perubahan Simpan aturan

**Pembahasan :** Memakai vm-server dan vm-client yang sudah dibuat kemarin semua juga sudah diset h type All ICMP – IPv4, kemudain server dibuat aturan baru yaitu TCP custom dengan rentang port 3000 khusus server

```
* Documentation: https://help.ubuntu.com
* Management:   https://landscape.canonical.com
* Support:       https://ubuntu.com/pro

System information as of Thu Oct 23 08:29:49 UTC 2025

System load: 0.0           Temperature: -273.1 C
Usage of /: 6.3% of 28.02GB Processes: 110
Memory usage: 22%         Users logged in: 0
Swap usage: 0%            IPv4 address for ens5: 172.31.24.75

* Ubuntu Pro delivers the most comprehensive open source security and
  compliance features.

https://ubuntu.com/aws/pro

Expanded Security Maintenance for Applications is not enabled.

0 updates can be applied immediately.

Enable ESM Apps to receive additional future security updates.
See https://ubuntu.com/esm or run: sudo pro status

The list of available updates is more than a week old.
To check for new updates run: sudo apt update

Last login: Thu Oct 16 08:43:01 2025 from 18.206.107.28
To run a command as administrator (user "root"), use "sudo <command>".
See "man sudo_root" for details.

ubuntu@ip-172-31-24-75:~$ sudo apt update && sudo apt upgrade -y
```

```
No VM guests are running outdated hypervisor (qemu) binaries on this host.
ubuntu@ip-172-31-24-75:~$ sudo apt install curl -y
Reading package lists... Done
Building dependency tree... Done
Reading state information... Done
curl is already the newest version (8.5.0-2ubuntu10.6).
curl set to manually installed.
0 upgraded, 0 newly installed, 0 to remove and 0 not upgraded.
ubuntu@ip-172-31-24-75:~$
```

**i-000a8ff1c561f54d3 (Vm-client)**

PublicIPs: 44.210.41.98 PrivateIPs: 172.31.24.75

Pembahasan : pengsintalan daftar paket dan pembauan sistem dengan perintah sudo apt update && sudo apt upgrade -y dan sudo apt install curl -y di server maupun clietn.

```

Enable ESM Apps to receive additional future security updates.
See https://ubuntu.com/esm or run: sudo pro status

*** System restart required ***
Last login: Thu Oct 23 08:29:36 2025 from 18.206.107.27
ubuntu@ip-172-31-19-37:~$ git clone https://github.com/akbarwjyy/student-api.git
Cloning into 'student-api'...
remote: Enumerating objects: 9, done.
remote: Counting objects: 100% (9/9), done.
remote: Compressing objects: 100% (6/6), done.
remote: Total 9 (delta 3), reused 8 (delta 2), pack-reused 0 (from 0)
Receiving objects: 100% (9/9), 9.20 KiB | 3.07 MiB/s, done.
Resolving deltas: 100% (3/3), done.
ubuntu@ip-172-31-19-37:~$ cd student-api
ubuntu@ip-172-31-19-37:~/student-api$ npm install

added 68 packages, and audited 69 packages in 2s

16 packages are looking for funding
  run `npm fund` for details

found 0 vulnerabilities
npm notice
npm notice New major version of npm available! 10.8.2 -> 11.6.2
npm notice Changelog: https://github.com/npm/cli/releases/tag/v11.6.2
npm notice To update run: npm install -g npm@11.6.2
npm notice
ubuntu@ip-172-31-19-37:~/student-api$ node.index.js
node.index.js: command not found
ubuntu@ip-172-31-19-37:~/student-api$ node index.js
Server berjalan di http://0.0.0.0:3000

```

Pembahasan : khusus server ini diinstal node js dan dihubungkan ke repository `curl -fsSL https://deb.nodesource.com/setup_20.x | sudo -E bash -` || `sudo apt install nodejs -y` || `git clone https://github.com/akbarwjyy/student-api.git` || `cd student-api` || `npm install` || `node index.js` agar server bisa serve seperti server pada umumnya

```
Diagnostics:
  The currently running kernel version is not the expected kernel version 6.14.0-1015-aws.

Restarting the system to load the new kernel will not be handled automatically, so you should consider rebooting.

Restarting services...
systemctl restart acpid.service chrony.service cron.service irqbalance.service multipathd.service packagekit.service polkit.service

Service restarts being deferred:
systemctl restart ModemManager.service
/etc/needrestart/restart.d/dbus.service
systemctl restart getty@tty1.service
systemctl restart networkd-dispatcher.service
systemctl restart serial-getty@ttyS0.service
systemctl restart systemd-logind.service
systemctl restart unattended-upgrades.service

No containers need to be restarted.

User sessions running outdated binaries:
ubuntu @ session #4: apt[1829], sshd[1001,1461]
ubuntu @ user manager service: systemd[1348]

No VM guests are running outdated hypervisor (qemu) binaries on this host.
ubuntu@ip-172-31-24-75:~$ sudo apt install curl -y
Reading package lists... Done
Building dependency tree... Done
Reading state information... Done
curl is already the newest version (8.5.0-2ubuntu10.6).
curl set to manually installed.
0 upgraded, 0 newly installed, 0 to remove and 0 not upgraded.
ubuntu@ip-172-31-24-75:~$ curl -X POST http://172.31.19.37:3000/register \-H "Content-Type: application/json" \-d '{"nama": "Irvan cahya", "nim": "245410034"}'
{"status": "success", "message": "Registrasi berhasil", "data": {"nama": "Irvan cahya", "nim": "245410034"}}ubuntu@ip-172-31-24-75:~$
```

**Kesimpulan :** Kembali ke virtual machine client untuk menambahkan data buatan dengan perintah `curl -X POST http://172.31.24.75:3000/register \-H "Content-Type: application/json" \-d '{"nama": "Irvan Cahya Nugraha", "nim": "245410034"}'` Pada ip 172.31.24.75:3000 sesuai dengan ip private server masing masing praktikan jika berhasil maka status sukses